浙江大学
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Recitation 3

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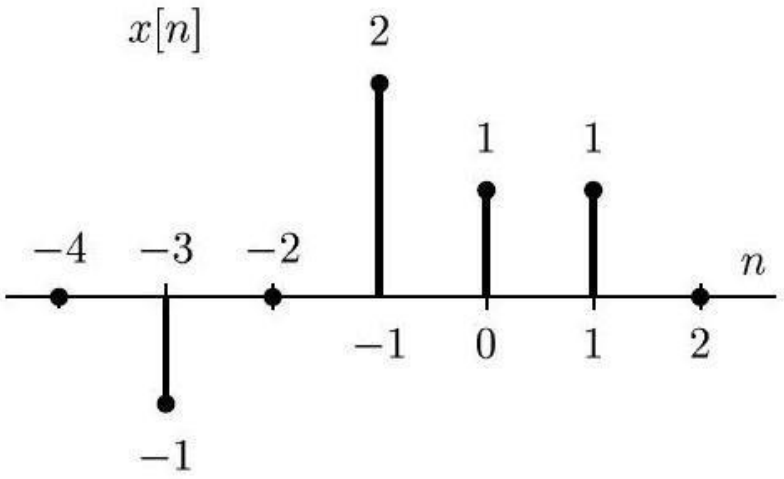
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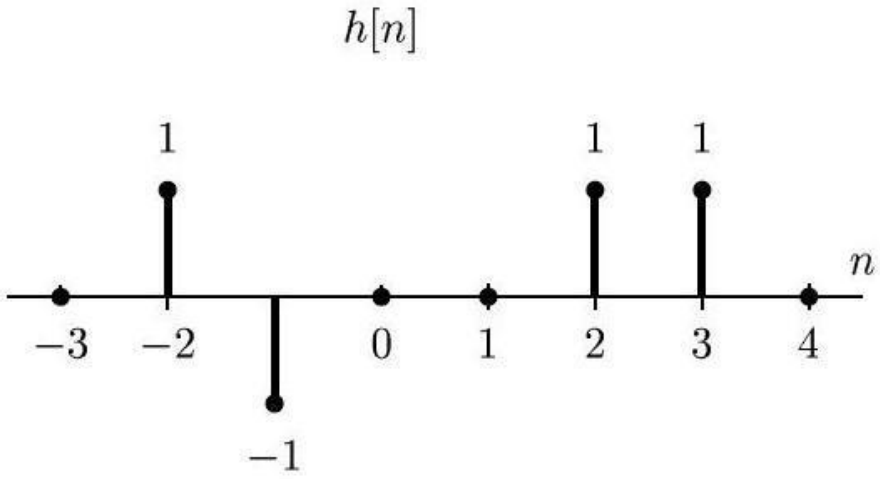


Problem 1

Compute the convolution $y[n] = x[n] * h[n]$ of each of the two following pairs of signals:

(a). $x[n]$ and $h[n]$ are depicted below

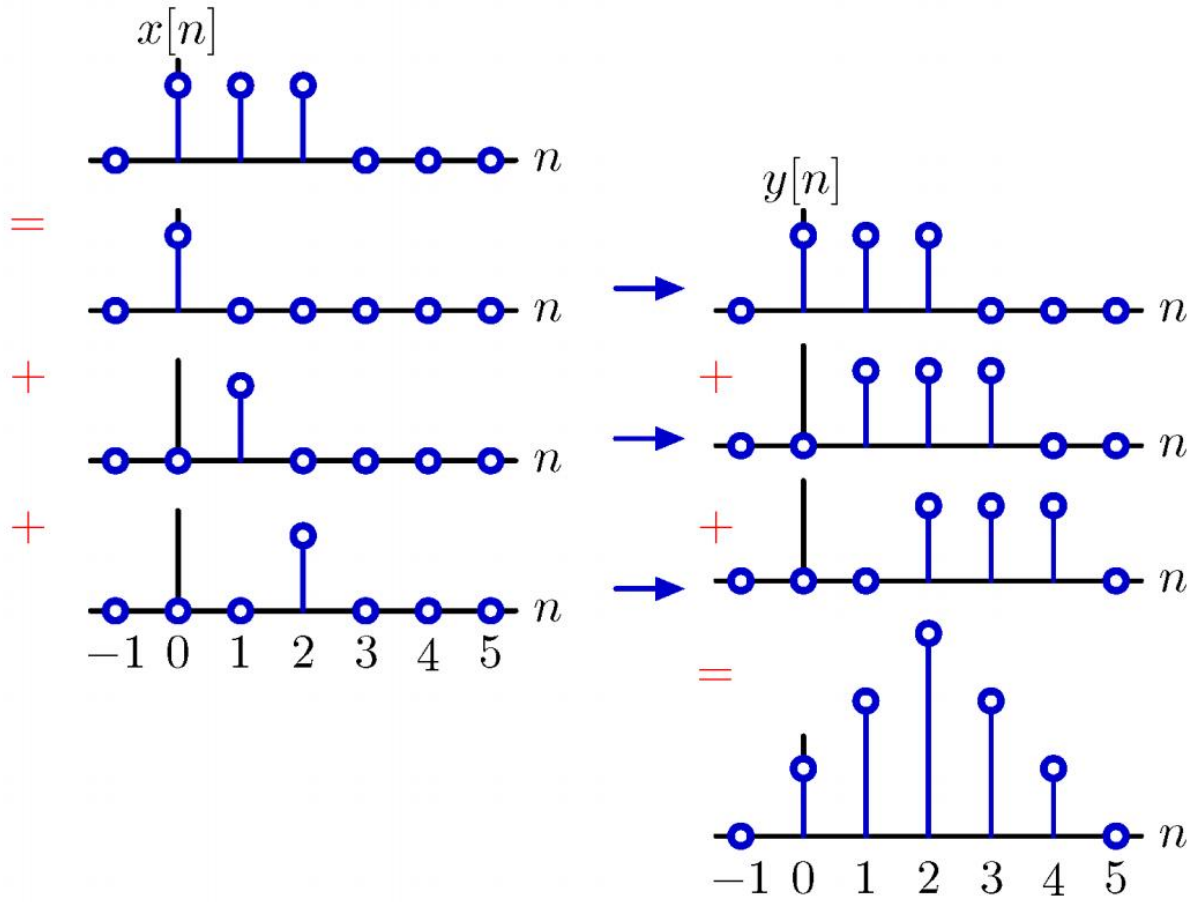




(b). $x[n] = u[n+4] - u[n-1]$, $h[n] = 2^n u[2-n]$.

Problem 1

Break input into additive parts and sum the responses to the parts.



Superposition is easy if the system is **linear** and **time-invariant**.

Problem 1

If a system is linear and time-invariant (LTI) then its output is the sum of weighted and shifted unit-sample responses.

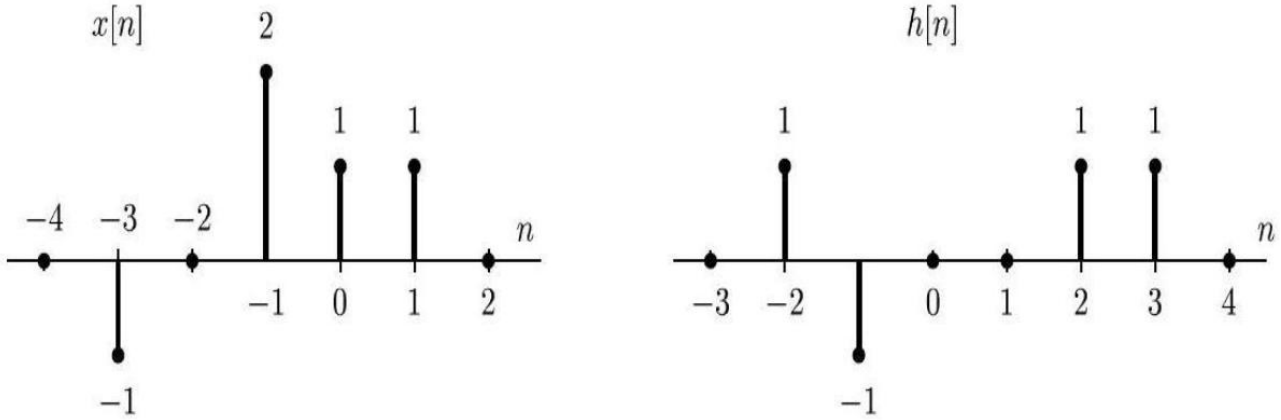
$$\delta[n] \rightarrow \boxed{\text{system}} \rightarrow h[n]$$

$$\delta[n-k] \rightarrow \boxed{\text{system}} \rightarrow h[n-k]$$

$$x[k]\delta[n-k] \rightarrow \boxed{\text{system}} \rightarrow x[k]h[n-k]$$

$$x[n] = \sum_{k=-\infty}^{\infty} x[k]\delta[n-k] \rightarrow \boxed{\text{system}} \rightarrow y[n] = \sum_{k=-\infty}^{\infty} x[k]h[n-k]$$

Problem 1 a



$$x[n] * h[n] = \sum_{k=-\infty}^{+\infty} x[k]\, h[n-k]$$

$\{1\ -1\ 0\ 0\ 1\ 1\}_{-2}$ $h[n]$

$\{-1\ 0\ 2\ 1\ 1\}_{-3}$ $x[n]$

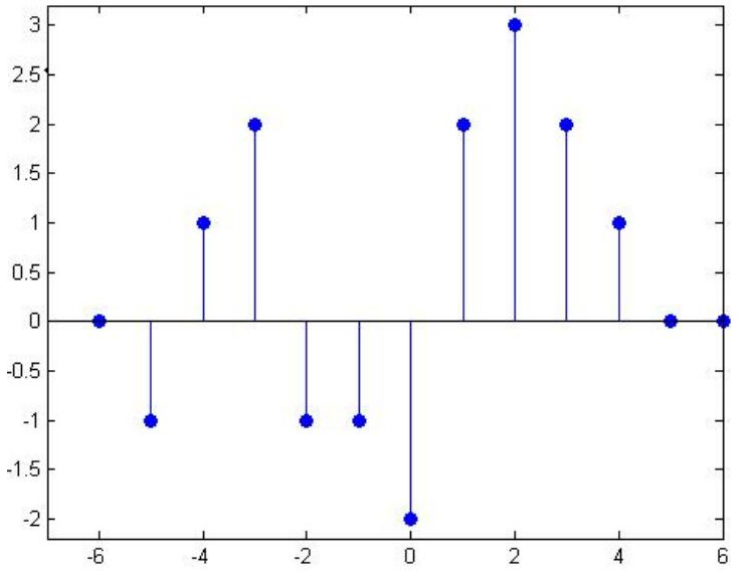
$-1\ 1\ 0\ 0\ -1\ -1$ $x[-3]\,h[n+3]$

$2\ -2\ 0\ 0\ 2\ 2$ $x[-1]\,h[n+1]$

$1\ -1\ 0\ 0\ 1\ 1$ $x[0]\,h[n]$

$1\ -1\ 0\ 0\ 1\ 1$ $x[1]\,h[n-1]$

$\{-1\ 1\ 2\ -1\ -1\ -2\ 2\ 3\ 2\ 1\}_{-5}$



Problem 1

两个离散序列的长度分别为M和N，则它们的卷积的长度L是多少？

设序列1的区间为[a, b]，序列2的区间为[c, d]，则两序列卷积的非零区间为[a+c, b+d]。

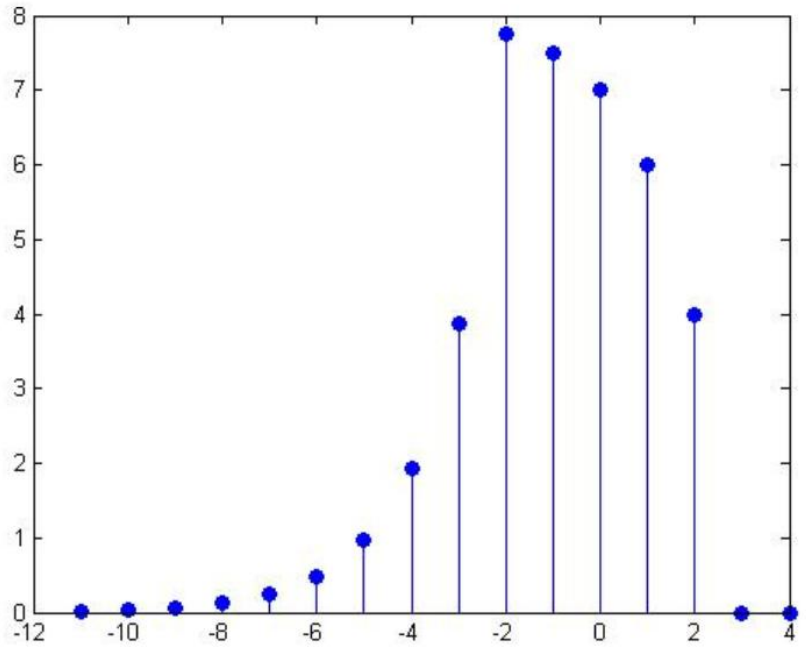
$$M = b - a + 1$$
$$N = d - c + 1$$

$$L = (b + d) - (a + c) + 1$$
$$= M + N - 1$$

Problem 1 b

(b). $x[n] = u[n+4] - u[n-1],\ h[n] = 2^n u[2-n].$

$$y[n] = x[n] * h[n] = \sum_{k=-\infty}^{2} 2^k u[2-k](u[n-k+4] - u[n-k-1]) = \sum_{k=n}^{n+4} 2^k u[2-k] = \begin{cases} 31\times 2^n, & n \le -2 \\ 7.5, & n = -1 \\ 7, & n = 0 \\ 6, & n = 1 \\ 4, & n = 2 \\ 0, & else \end{cases}$$

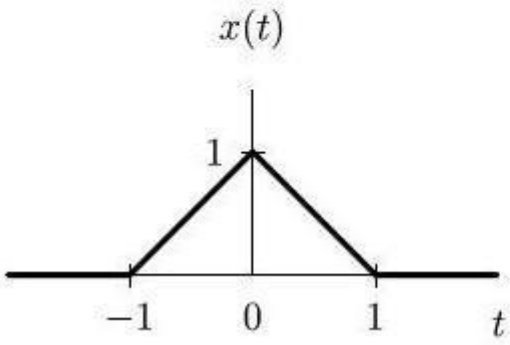


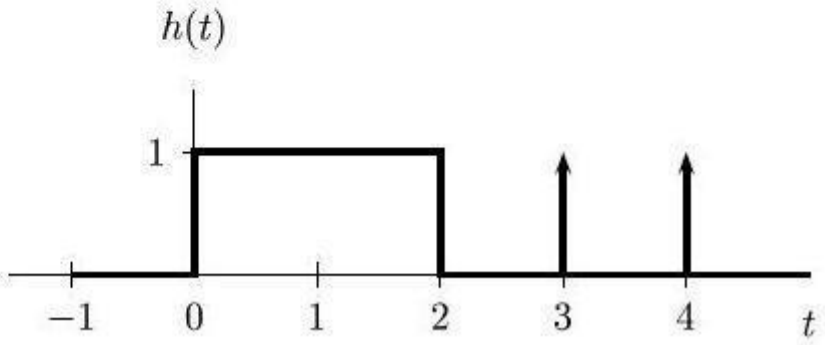
Problem 2

Compute the convolution $y(t) = x(t) * h(t)$ for each of the following pairs of signals:

(a). $x(t) = e^{-t}u(t+1)$, $h(t) = e^{2t}u(-t)$

(b). $x(t)$ and $h(t)$ are depicted below:





Problem 2 a

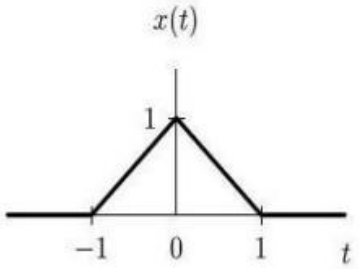
(a). $x(t) = e^{-t}u(t+1),\ h(t) = e^{2t}u(-t)$

$$y(t) = \int_{-\infty}^{\infty} e^{-\tau}u(\tau+1)e^{2t-2\tau}u(\tau-t)d\tau$$

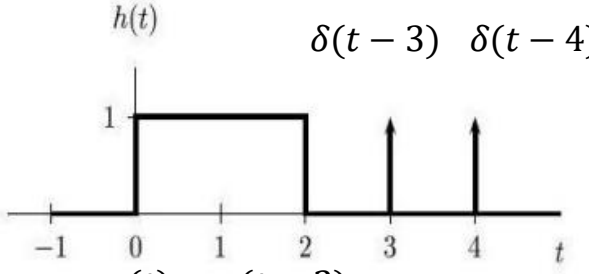
$$= \begin{cases} e^{2t}\int_{t}^{\infty} e^{-3\tau}\,d\tau, t > -1 \\ e^{2t}\int_{-1}^{\infty} e^{-3\tau}\,d\tau, t \leq -1 \end{cases}$$

$$y(t) = \begin{cases} \frac{1}{3}e^{-t}, & t > -1 \\ \frac{1}{3}e^{2t+3}, & t \leq -1 \end{cases}$$

Problem 2 b



*



$\delta(t-3)\quad \delta(t-4)$

$u(t)-u(t-2)$

$$y(t)=x(t-3)+x(t-4)+\int_{t-2}^{t}x(\tau)d\tau$$

$$=\begin{cases}\frac{1}{2}(1+t)^2, & -1\le t<0\\ -\frac{1}{2}t^2+t+\frac{1}{2}, & 0\le t<2\\ \frac{1}{2}t^2-2t+\frac{5}{2}, & 2\le t<3\\ 1, & 3\le t<4\\ -t+5, & 4\le t<5\\ 0, & else\end{cases}$$

Problem 3

The following are impulse responses of either discrete-time or continuous-time LTI systems. Determine whether each system is causal and/or stable. Justify your answer:

(a). $h[n] = 2^n u[3-n]$

(b). $h(t) = u(1-t) - \frac{1}{2}e^{-t}u(t)$

(c). $h[n] = [1-(0.99)^n]u[n]$

(d). $h(t) = e^{15t}\,[u(t-1) - u(t-100)]$

CT LTI

Causal: $h(t) = 0,\ t < 0 \Leftrightarrow$ ROC is right side

Stable: $\int_{-\infty}^{\infty} |h(t)|\, dt < \infty \Leftrightarrow$ ROC includes jw axis

DT LTI

Causal: $h[n] = 0,\ n < 0 \Leftrightarrow$ ROC includes ∞

Stable: $\sum_{n=-\infty}^{\infty} |h[n]| < \infty \Leftrightarrow$ ROC includes unit circle

Problem 3

The following are impulse responses of either discrete-time or continuous-time LTI systems. Determine whether each system is causal and/or stable. Justify your answer:

(a). $h[n] = 2^n u[3-n]$

(b). $h(t) = u(1-t) - \frac{1}{2}e^{-t}u(t)$

(c). $h[n] = [1-(0.99)^n]u[n]$

(d). $h(t) = e^{15t}\,[u(t-1) - u(t-100)]$

(a)noncausal， stable

$$\sum_{k=-\infty}^{+\infty} |h[k]| = \sum_{k=-\infty}^{3} 2^k = 16$$

(c)causal， unstable

$$\sum_{k=-\infty}^{+\infty} |h[k]| = \sum_{k=0}^{+\infty} 1-(0.99)^k = \infty$$

(b)noncausal， unstable

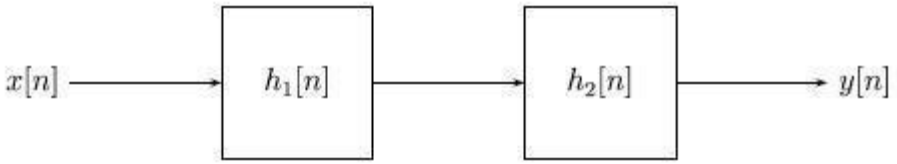
$$\int_{-\infty}^{+\infty} |h(\tau)|d\tau \geq \int_{-\infty}^{0} |h(\tau)|d\tau = \int_{-\infty}^{0} 1 d\tau = \infty$$

(d)causal, stable

$$\int_{-\infty}^{+\infty} |h(\tau)|d\tau = \int_{1}^{99} e^{15\tau} d\tau < \infty$$

Problem 4

Consider the cascade of LTI systems with unit sample responses $h_1[n]$ and $h_2[n]$ depicted below:

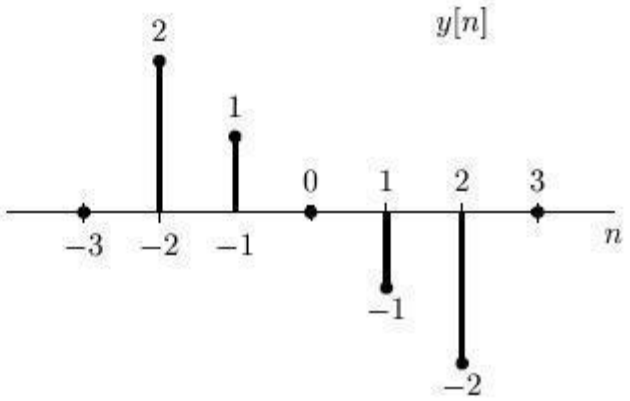


Suppose we are given the following information:

- $h_2[n] = \delta[n] - \delta[n-1]$
- If the input is

$$x[n] = u[n] - u[n-2]$$

then the output is as depicted below



Find $h_1[n]$.

Problem 4

Method 1

$$y[n]=x[n]*h_1[n]*h_2[n] \xrightarrow{L} H_1=\frac{Y}{X\cdot H_2}$$

$$\begin{cases} h_2[n]=\delta[n]-\delta[n-1] & \xrightarrow{L} \quad H_2=1-z^{-1} \\ x[n]=u[n]-u[n-2] & \xrightarrow{L} \quad X=\dfrac{1}{1-z^{-1}}-\dfrac{z^{-2}}{1-z^{-1}} \\ y[n]=2\delta[n+2]+\delta[n+1]-\delta[n-1]-2\delta[n-2] & \xrightarrow{L} \quad Y=2z^2+z-z^{-1}-2z^{-2} \end{cases}$$

$$\Rightarrow H_1=\frac{Y}{X\cdot H_2}=2z^2+z+2$$

Therefore, we have $h_1[n]=2\delta[n+2]+\delta[n+1]+2\delta[n]$.

Problem 4

Method 2

$$y[n] = x[n] * h_1[n] * h_2[n]$$

$$x[n] * h_2[n] = (u[n] - u[n-2]) * (\delta[n] - \delta[n-1]) = (\delta[n] - \delta[n-2])$$

$$y[n] = (\delta[n] - \delta[n-2]) * h_1[n] = h_1[n] - h_1[n-2]$$

$$y[n] = 2\delta[n+2] + \delta[n+1] - \delta[n-1] - 2\delta[n-2]$$

$y[n] = h_1[n] - h_1[n-2]$

$h_1[n]$	2	1	2		
$h_1[n-2]$			2	1	2
$y[n]$	2	1	0	-1	-2

Problem 5

6.1. Consider a continuous-time LTI system with frequency response $H(j\omega) = |H(j\omega)|\, e^{j\measuredangle H(j\omega)}$ and real impulse response $h(t)$. Suppose that we apply an input $x(t) = \cos(\omega_0 t + \phi_0)$ to this system. The resulting output can be shown to be of the form

$$y(t) = Ax(t - t_0),$$

where A is a nonnegative real number representing an *amplitude-scaling* factor and t_0 is a time delay.

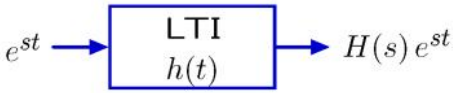
(a) Express A in terms of $|H(j\omega_0)|$.

(b) Express t_0 in terms of $\measuredangle H(j\omega_0)$.

Problem 5

If $x(t) = e^{st}$ and $h(t)$ is the impulse response then

$$y(t) = (h * x)(t) = \int_{-\infty}^{\infty} h(\tau) e^{s(t-\tau)} d\tau = e^{st} \int_{-\infty}^{\infty} h(\tau) e^{-s\tau} d\tau = H(s)\, e^{st}$$



Let $x(t) = \cos \omega_0 t$ (for all time), which can be written as

$$x(t) = \frac{1}{2}\left(e^{j\omega_0 t} + e^{-j\omega_0 t}\right)$$

The response to a sum is the sum of the responses,

$$\begin{aligned} y(t) &= \frac{1}{2}\left(H(j\omega_0)e^{j\omega_0 t} + H(-j\omega_0)e^{-j\omega_0 t}\right) \\ &= \mathrm{Re}\left\{H(j\omega_0)e^{j\omega_0 t}\right\} \\ &= \mathrm{Re}\left\{|H(j\omega_0)|e^{j\angle H(j\omega_0)}e^{j\omega_0 t}\right\} \\ &= |H(j\omega_0)|\mathrm{Re}\left\{e^{j\omega_0 t + j\angle H(j\omega_0)}\right\} \end{aligned}$$

$$y(t) = |H(j\omega_0)| \cos\left(\omega_0 t + \angle\left(H(j\omega_0)\right)\right).$$

Problem 5

6.1. Consider a continuous-time LTI system with frequency response $H(j\omega) = |H(j\omega)|\, e^{j\measuredangle H(j\omega)}$ and real impulse response $h(t)$. Suppose that we apply an input $x(t) = \cos(\omega_0 t + \phi_0)$ to this system. The resulting output can be shown to be of the form

$$y(t) = Ax(t - t_0),$$

where A is a nonnegative real number representing an *amplitude-scaling* factor and t_0 is a time delay.

(a) Express A in terms of $|H(j\omega_0)|$.

(b) Express t_0 in terms of $\measuredangle H(j\omega_0)$.

$$x(t) = \cos\left(\omega_0\left(t + \frac{\phi_0}{\omega_0}\right)\right)$$

$$y(t) = |H(j\omega_0)|\cos\left(\omega_0\left(t + \frac{\phi_0}{\omega_0}\right) + \angle H(j\omega_0)\right)$$

$$= |H(j\omega_0)|\cos\left(\omega_0\left(t + \frac{\phi_0}{\omega_0} + \frac{\angle H(j\omega_0)}{\omega_0}\right)\right)$$

$$A = |H(j\omega_0)|$$

$$t_0 = -\frac{\angle H(j\omega_0)}{\omega_0}$$

Problem 6

6.3. Consider the following frequency response for a causal and stable LTI system:

$$H(j\omega) = \frac{1 - j\omega}{1 + j\omega}.$$

(a) Show that $|H(j\omega)| = A$, and determine the value of A.

(b) Determine which of the following statements is true about $\tau(\omega)$, the group delay of the system. (*Note:* $\tau(\omega) = -d(\measuredangle H(j\omega))/d\omega$, where $\measuredangle H(j\omega)$ is expressed in a form that does not contain any discontinuities.)

1. $\tau(\omega) = 0$ for $\omega > 0$
2. $\tau(\omega) > 0$ for $\omega > 0$
3. $\tau(\omega) < 0$ for $\omega > 0$

$$|H(j\omega)| = \frac{|1-j\omega|}{|1+j\omega|} = \frac{\sqrt{1+\omega^2}}{\sqrt{1+\omega^2}} = 1$$

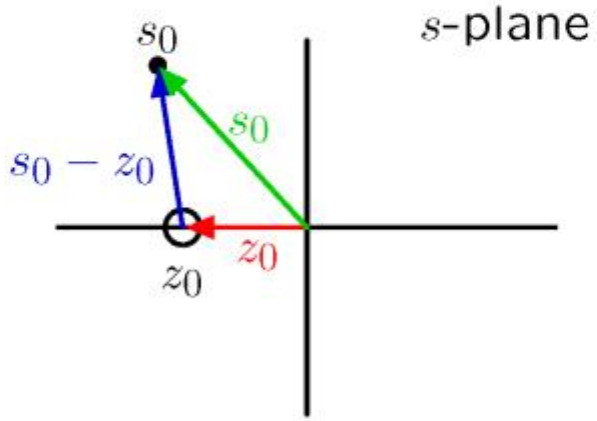
$$H(j\omega) = \frac{(1-j\omega)^2}{(1+j\omega)(1-j\omega)} = \frac{1-\omega^2-2j\omega}{1+\omega^2}$$

$$\measuredangle H(j\omega) = \arctan\frac{-2\omega}{1-\omega^2}$$

$$\tau(\omega) = \frac{2}{1+\omega^2}$$

2 is true

$$H(s_0) = K\frac{(s_0 - z_0)(s_0 - z_1)(s_0 - z_2)\cdots}{(s_0 - p_0)(s_0 - p_1)(s_0 - p_2)\cdots}$$



在对数坐标下（波特图），幅值是对数形式：

$$20\log|H(j\omega)|$$

当 ω 远小于任何零点和极点时，系统的增益近似于常数 $20\log K$。

当频率增加并接近某个零点或极点时，它们会改变系统的斜率：

- **零点（z_i）：** 使幅值曲线上升，增加 20 dB/dec（十倍频程）。
- **极点（p_i）：** 使幅值曲线下降，减少 20 dB/dec（十倍频程）。

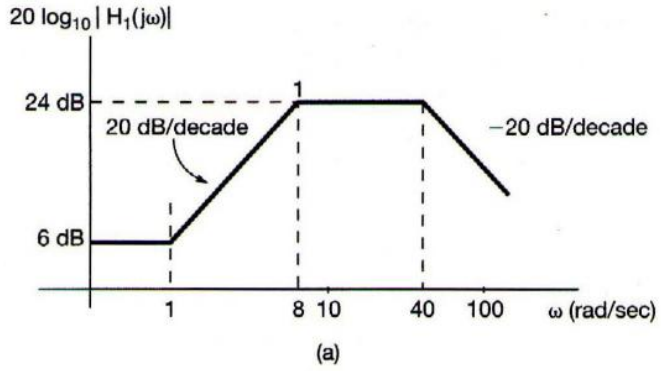
如果有二**阶极点**，则下降率是 40 dB/dec，如果有二**阶零点**，则上升率是 40 dB/dec。

相频图的相位变化不像幅频图那样严格在零极点位置拐折，而是一个**缓慢过渡**：

- **零点（Zero）** 使相位**变化 ±90°**，但相位变化在 $0.1z$ 到 $10z$ 之间分布，而不是严格在 z 处突变。
- **极点（Pole）** 使相位**变化 ±90°**，但相位变化在 $0.1p$ 到 $10p$ 之间分布，而不是严格在 p 处突变。
- LHP 零点：$+90^\circ$ 相位上升。
- LHP 极点：-90° 相位下降。
- RHP 零点：-90° 相位下降（与 LHP 零点相反）。
- RHP 极点：一般不考虑（因果系统不允许）。
- 相位的变化幅度取决于**零极点的阶数**：
  - **单个零点/极点：变化** $\pm 90^\circ$。
  - **二阶零点/极点：变化** $\pm 180^\circ$。

Problem 7

6.12. A continuous-time LTI system S with frequency response $H(j\omega)$ is constructed by cascading two continuous-time LTI systems with frequency responses $H_1(j\omega)$ and $H_2(j\omega)$, respectively. Figures P6.12(a) and P6.12(b) show the straight-line approximations of the Bode magnitude plots of $H_1(j\omega)$ and $H(j\omega)$, respectively. Specify $H_2(j\omega)$.



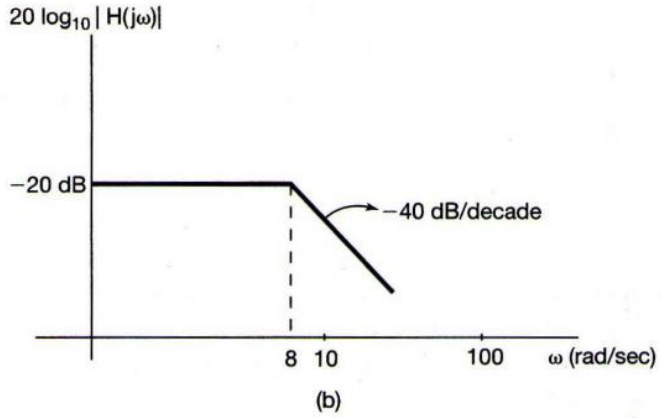


Figure P6.12

$$z_1 = -1$$

$$p_1 = -8, \ -40$$

$$H_1(s) = A\frac{s+1}{(s+8)(s+40)}$$

$$H_1(j\omega) = A\frac{j\omega+1}{(j\omega+8)(j\omega+40)}$$

$$H_1(j0) = A\frac{1}{8\times 40} = \frac{A}{320} = 2$$

$$A = 640$$

$$H_1(j\omega) = 640\frac{j\omega+1}{(j\omega+8)(j\omega+40)}$$

$$H(s) = B\frac{1}{(s+8)^2}$$

$$H(j\omega) = B\frac{1}{(j\omega+8)^2}$$

$$H(j0) = \frac{B}{64} = 0.1$$

$$B = 6.4$$

$$H(j\omega) = \frac{6.4}{(j\omega+8)^2}$$

$$p = -8$$

$$H_2(j\omega) = \frac{H(j\omega)}{H_1(j\omega)} = \frac{0.01(j\omega+40)}{(j\omega+1)(j\omega+8)}$$

只考虑因果稳定系统

Problem 8

6.21. A causal LTI filter has the frequency response $H(j\omega)$ shown in Figure P6.21. For each of the input signals given below, determine the filtered output signal $y(t)$.

(a) $x(t) = e^{jt}$ **(b)** $x(t) = (\sin \omega_0 t)u(t)$

(c) $X(j\omega) = \frac{1}{(j\omega)(6+j\omega)}$ **(d)** $X(j\omega) = \frac{1}{2+j\omega}$

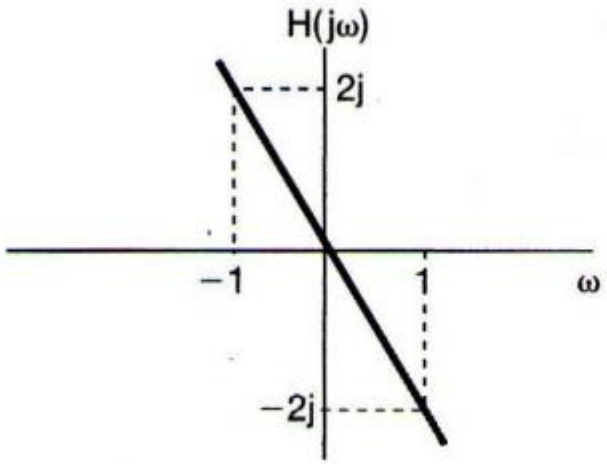


Figure P6.21

Problem 8 a

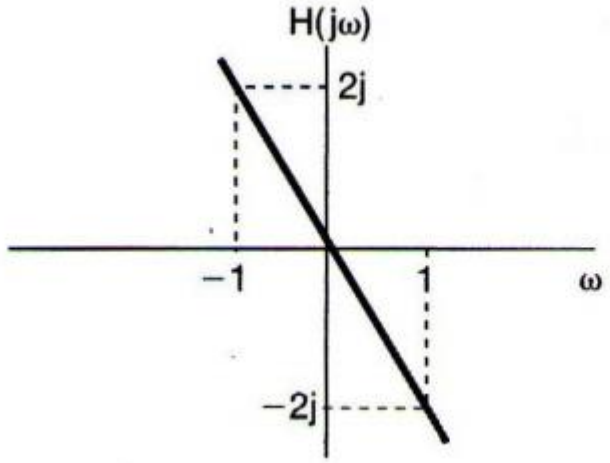


Figure P6.21

(a) $x(t) = e^{jt}$ **(b)** $x(t) = (\sin \omega_0 t)u(t)$

(c) $X(j\omega) = \frac{1}{(j\omega)(6+j\omega)}$ **(d)** $X(j\omega) = \frac{1}{2+j\omega}$

$H(j\omega) = -2j\omega$

$Y(j\omega) = X(j\omega)H(j\omega) = -2j\omega X(j\omega)$

$Y(s) = -2sX(s)$

$y(t) = -2\frac{dx(t)}{dt}$

(a) $y(t) = -2je^{jt}$

(b) $y(t) = -2\omega_0 \cos(\omega_0 t)u(t)$

(c) $X(s) = \frac{1}{s(s+6)}$

$Y(s) = \frac{-2}{6+s}$

$y(t) = -2e^{-6t}u(t)$

(d) $x(t) = e^{-2t}u(t)$

$y(t) = 4e^{-2t}u(t) - 2\delta(t)$

Problem 9

6.28. **(a)** Sketch the Bode plots for the following frequency responses:

(i) $1 + (j\omega/10)$

(ii) $1 - (j\omega/10)$

(iii) $\frac{16}{(j\omega+2)^4}$

(iv) $\frac{1-(j\omega/10)}{1+j\omega}$

(v) $\frac{(j\omega/10)-1}{1+j\omega}$

(vi) $\frac{1+(j\omega/10)}{1+j\omega}$

(vii) $\frac{1-(j\omega/10)}{(j\omega)^2+(j\omega)+1}$

(viii) $\frac{10+5j\omega+10(j\omega)^2}{1+(j\omega/10)}$

(ix) $1 + j\omega + (j\omega)^2$

(x) $1 - j\omega + (j\omega)^2$

(xi) $\frac{(j\omega+10)(10j\omega+1)}{[(j\omega/100+1)][((j\omega)^2+j\omega+1)]}$

在对数坐标下（波特图），幅值是对数形式：

$$20\log|H(j\omega)|$$

当 ω 远小于任何零点和极点时，系统的增益近似于常数 $20\log K$。

当频率增加并接近某个零点或极点时，它们会改变系统的斜率：

- **零点（z_i）：** 使幅值曲线上升，增加 20 dB/dec（十倍频程）。
- **极点（p_i）：** 使幅值曲线下降，减少 20 dB/dec（十倍频程）。

如果有二**阶极点**，则下降率是 40 dB/dec，如果有二**阶零点**，则上升率是 40 dB/dec。

相频图的相位变化不像幅频图那样严格在零极点位置拐折，而是一个**缓慢过渡**：

- **零点（Zero）** 使相位**变化 ±90°**，但相位变化在 $0.1z$ 到 $10z$ 之间分布，而不是严格在 z 处突变。
- **极点（Pole）** 使相位**变化 ±90°**，但相位变化在 $0.1p$ 到 $10p$ 之间分布，而不是严格在 p 处突变。
- LHP 零点：$+90°$ 相位上升。
- LHP 极点：$-90°$ 相位下降。
- RHP 零点：$-90°$ 相位下降（与 LHP 零点相反）。
- RHP 极点：一般不考虑（因果系统不允许）。
- 相位的变化幅度取决于**零极点的阶数**：
  - **单个零点/极点：变化** $\pm 90°$。
  - **二阶零点/极点：变化** $\pm 180°$。

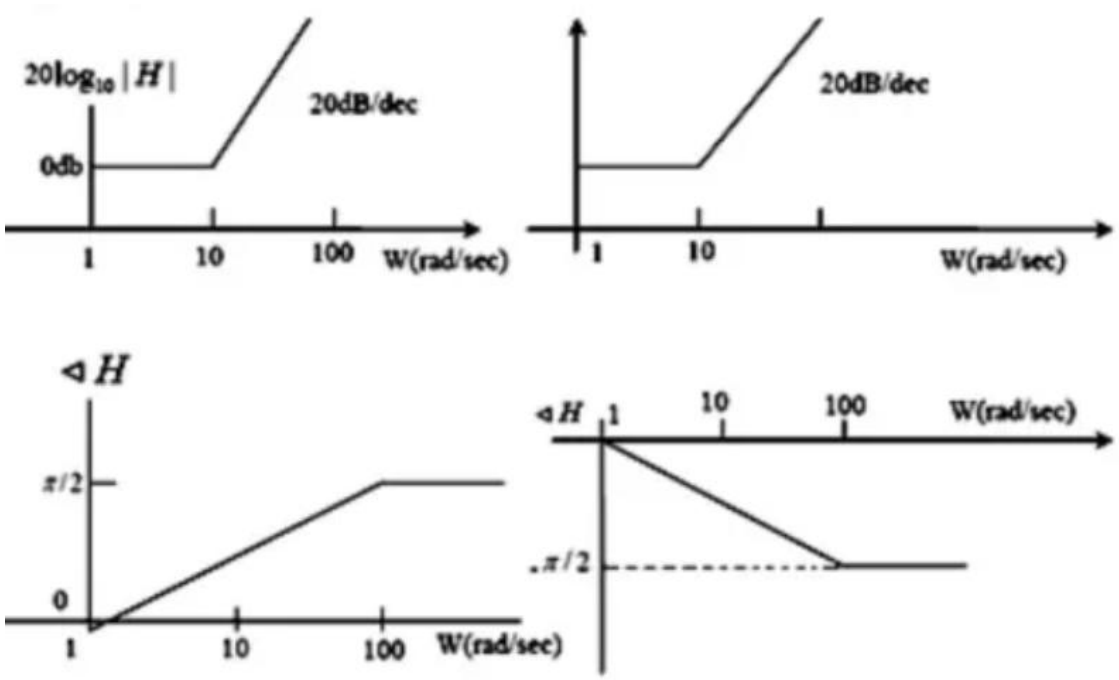
Problem 9

(i) $1 + (j\omega/10)$

$z_0 = -10$

(ii) $1 - (j\omega/10)$

$z_0 = 10$

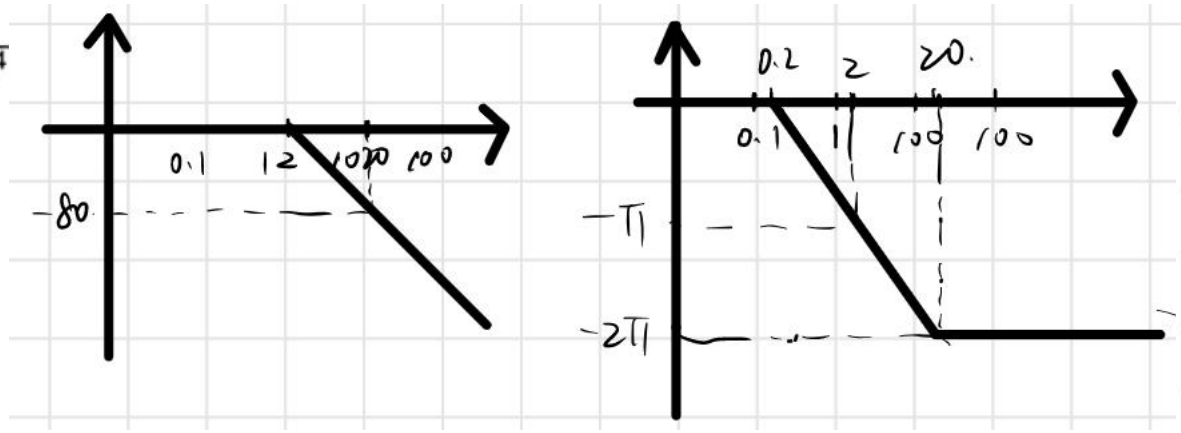


- **零点（z_i）：** 使幅值曲线上升，增加 20 dB/dec（十倍频程）。
- **极点（p_i）：** 使幅值曲线下降，减少 20 dB/dec（十倍频程）。
  - LHP 零点：$+90^\circ$ 相位上升。
  - LHP 极点：-90° 相位下降。
  - RHP 零点：-90° 相位下降（与 LHP 零点相反）。
  - RHP 极点：一般不考虑（因果系统不允许）。

Problem 9

(iii) $\frac{16}{(j\omega+2)^4}$

$p_0 = -2$

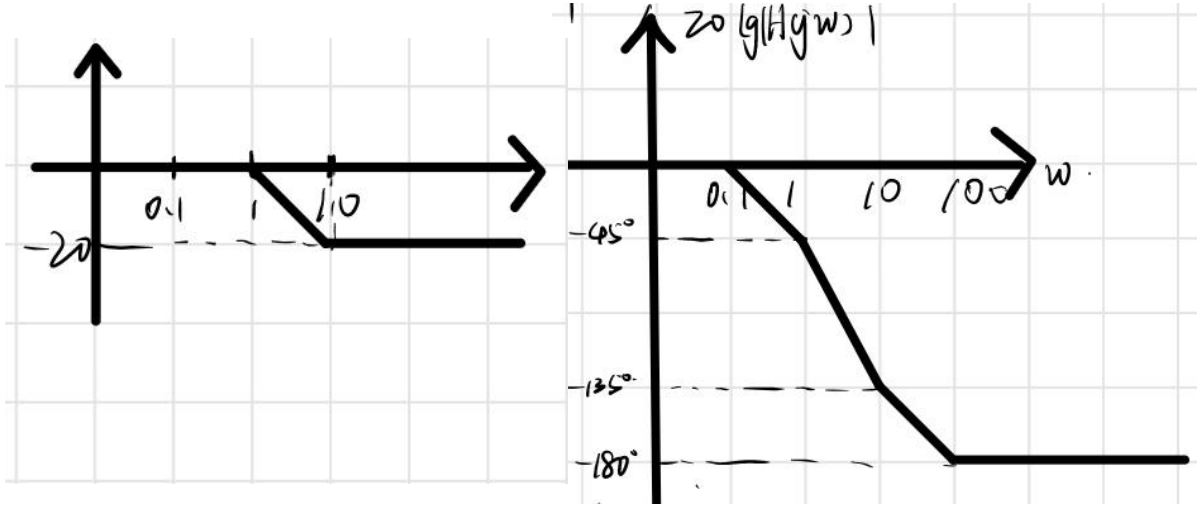


- **零点（z_i）：** 使幅值曲线上升，增加 20 dB/dec（十倍频程）。
- **极点（p_i）：** 使幅值曲线下降，减少 20 dB/dec（十倍频程）。
- LHP 零点：$+90°$ 相位上升。
- LHP 极点：$-90°$ 相位下降。
- RHP 零点：$-90°$ 相位下降（与 LHP 零点相反）。
- RHP 极点：一般不考虑（因果系统不允许）。

(iv) $\frac{1-(j\omega/10)}{1+j\omega}$

$z_0 = 10$

$p_0 = -1$

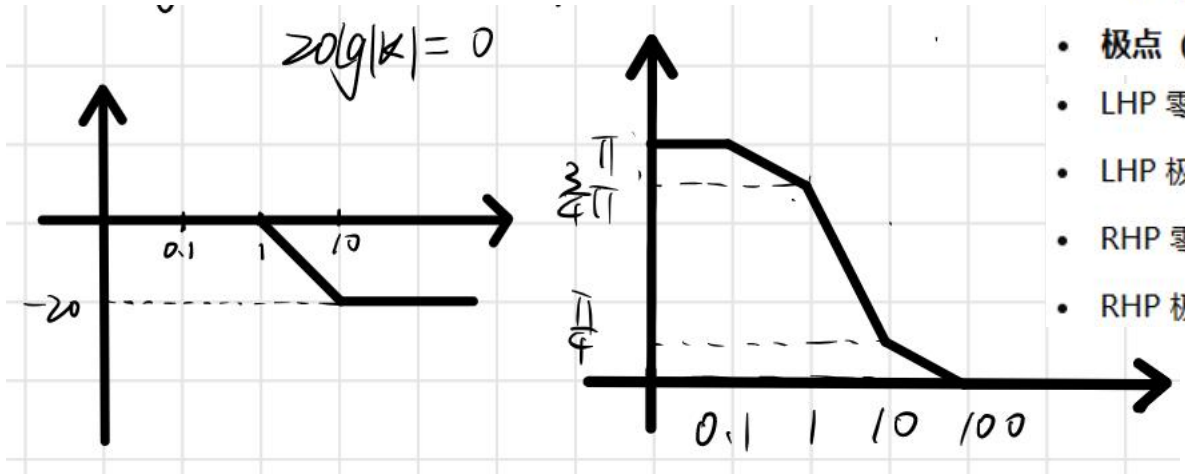


Problem 9

(v) $\dfrac{(j\omega/10)-1}{1+j\omega}$

$z_0 = 10$

$p_0 = -1$

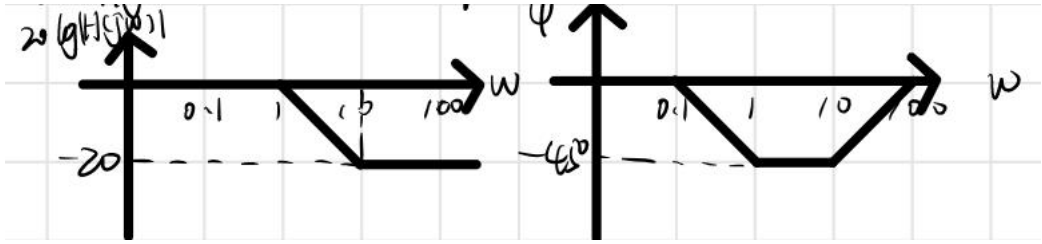


- **零点（z_i）：** 使幅值曲线上升，增加 20 dB/dec（十倍频程）。
- **极点（p_i）：** 使幅值曲线下降，减少 20 dB/dec（十倍频程）。
- LHP 零点：$+90°$ 相位上升。
- LHP 极点：$-90°$ 相位下降。
- RHP 零点：$-90°$ 相位下降（与 LHP 零点相反）。
- RHP 极点：一般不考虑（因果系统不允许）。

(vi) $\dfrac{1+(j\omega/10)}{1+j\omega}$

$z_0 = -10$

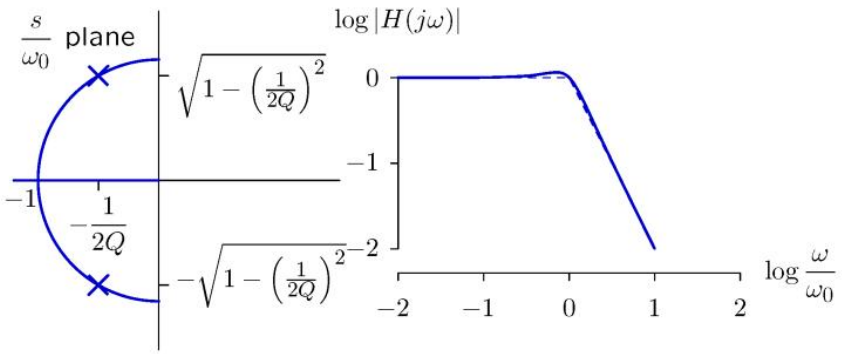
$p_0 = -1$



Problem 9

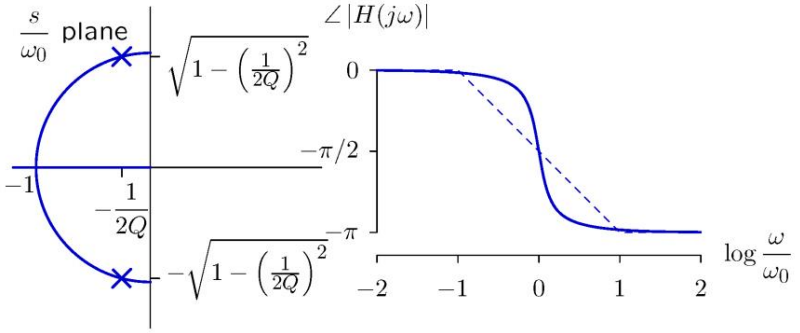
The frequency-response magnitude of a high-Q system is peaked.

$$H(s) = \frac{1}{1 + \frac{1}{Q}\frac{s}{\omega_0} + \left(\frac{s}{\omega_0}\right)^2}$$



As Q increases, the phase changes more abruptly with ω.

$$H(s) = \frac{1}{1 + \frac{1}{Q}\frac{s}{\omega_0} + \left(\frac{s}{\omega_0}\right)^2}$$



$$H(j\omega) = \frac{1}{1 - \left(\frac{\omega}{\omega_0}\right)^2 + j\frac{\omega}{Q\omega_0}}. \qquad \angle H(j\omega) = -\arctan\left(\frac{\text{虚部}}{\text{实部}}\right) = -\arctan\left(\frac{\frac{\omega}{Q\omega_0}}{1 - \left(\frac{\omega}{\omega_0}\right)^2}\right).$$

$$\left.\frac{d(\angle H(j\omega))}{d\omega}\right|_{\omega=\omega_0} = -\frac{2Q}{\omega_0},$$

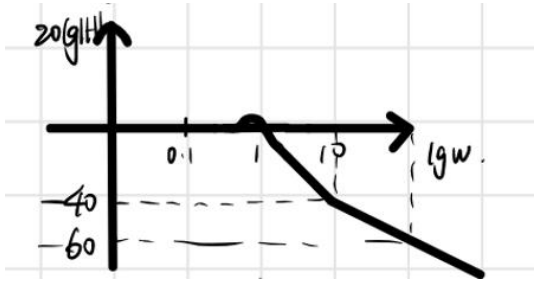
Problem 9

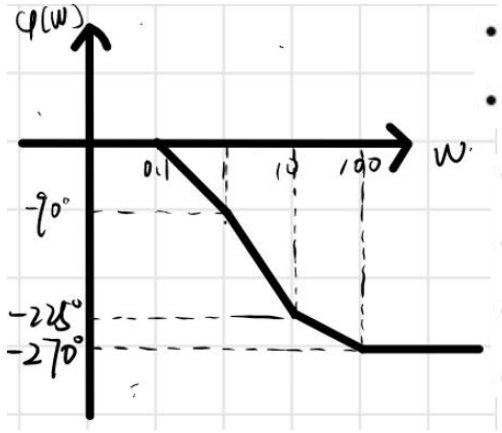
(vii) $\dfrac{1-(j\omega/10)}{(j\omega)^2+(j\omega)+1}$

$z = 10$

$\mathrm{Re}\{p\} = -\dfrac{1}{2}$

$\omega_0 = 1$





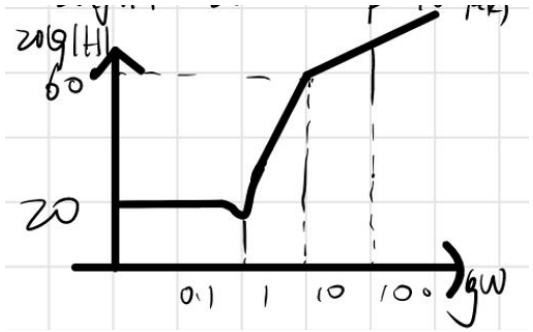
- **零点（z_i）：** 使幅值曲线上升，增加 20 dB/dec（十倍频程）。
- **极点（p_i）：** 使幅值曲线下降，减少 20 dB/dec（十倍频程）。
- LHP 零点：$+90°$ 相位上升。
- LHP 极点：$-90°$ 相位下降。
- RHP 零点：$-90°$ 相位下降（与 LHP 零点相反）。
- RHP 极点：一般不考虑（因果系统不允许）。

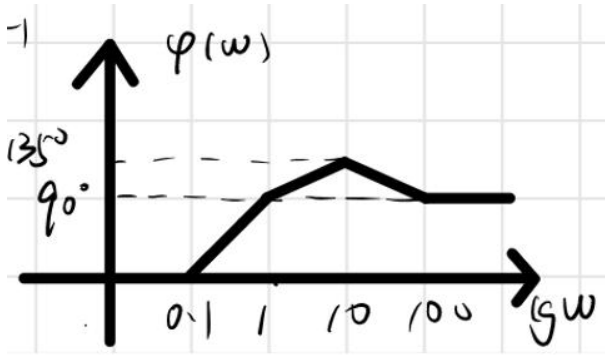
(viii) $\dfrac{10+5j\omega+10(j\omega)^2}{1+(j\omega/10)}$

$\mathrm{Re}\{z\} = -\dfrac{1}{4}$

$\omega_0 = 1$

$p = -10$





Problem 9

(ix) $1 + j\omega + (j\omega)^2$

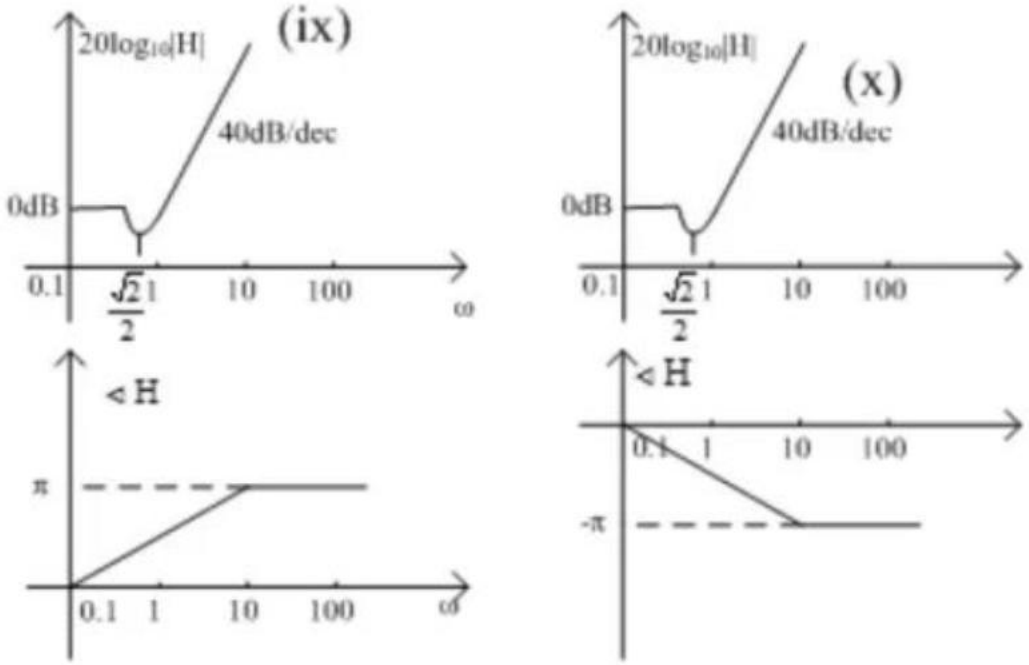
$\mathrm{Re}\{z\} = -\frac{1}{2}$

$\omega_0 = 1$

(x) $1 - j\omega + (j\omega)^2$

$\mathrm{Re}\{z\} = \frac{1}{2}$

$\omega_0 = 1$



- **零点** (z_i) **：** 使幅值曲线上升，增加 20 dB/dec（十倍频程）。
- **极点** (p_i) **：** 使幅值曲线下降，减少 20 dB/dec（十倍频程）。
  - LHP 零点：$+90^\circ$ 相位上升。
  - LHP 极点：-90° 相位下降。
  - RHP 零点：-90° 相位下降（与 LHP 零点相反）。
  - RHP 极点：一般不考虑（因果系统不允许）。

Problem 9

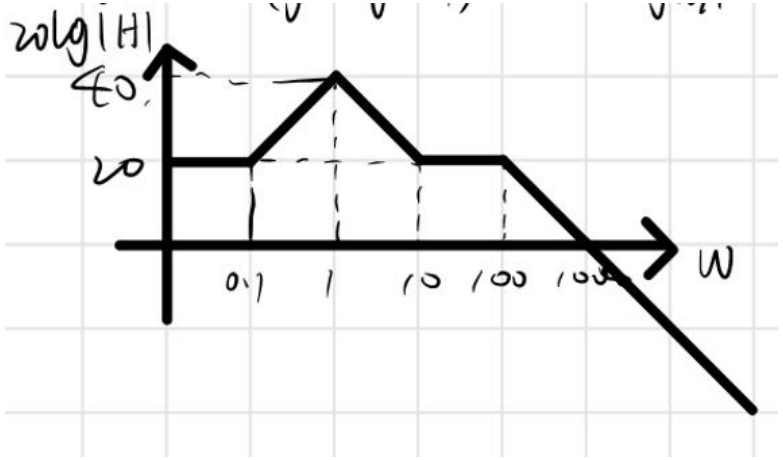
(xi) $$\frac{(j\omega+10)(10j\omega+1)}{[(j\omega/100+1)][((j\omega)^2+j\omega+1)]}$$

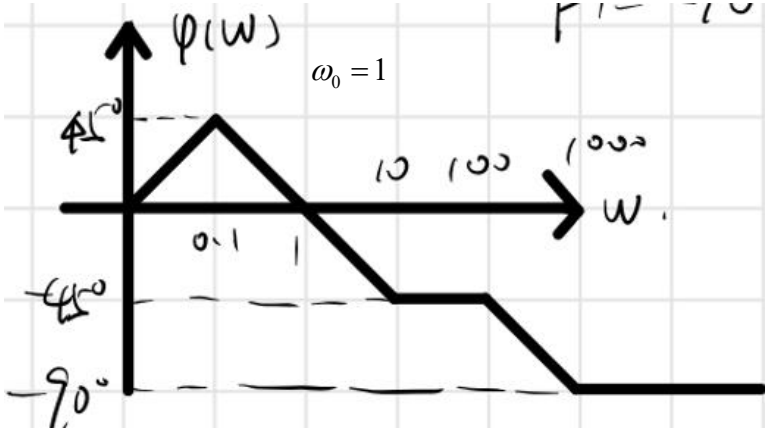
$z_0 = -10,\ z_1 = -0.1$

$p_0 = -100,\ \mathrm{Re}\{p_1\} = -\frac{1}{2}$

$\omega_0 = 1$

- **零点** (z_i)：使幅值曲线上升，增加 20 dB/dec（十倍频程）。
- **极点** (p_i)：使幅值曲线下降，减少 20 dB/dec（十倍频程）。
  - LHP 零点：$+90°$ 相位上升。
  - LHP 极点：$-90°$ 相位下降。
  - RHP 零点：$-90°$ 相位下降（与 LHP 零点相反）。
  - RHP 极点：一般不考虑（因果系统不允许）。





Problem 10 a

6.37. Consider a causal LTI system whose frequency response is given as:

$$H(e^{j\omega}) = e^{-j\omega}\frac{1-\frac{1}{2}e^{j\omega}}{1-\frac{1}{2}e^{-j\omega}}.$$

(a) Show that $|H(e^{j\omega})|$ is unity at all frequencies.

(b) Show that

$$\measuredangle H(e^{j\omega}) = -\omega - 2\tan^{-1}\left(\frac{\frac{1}{2}\sin\omega}{1-\frac{1}{2}\cos\omega}\right).$$

$$H(e^{j\omega}) = \frac{e^{-j\omega}-\frac{1}{2}}{1-\frac{1}{2}e^{-j\omega}}$$

$$= \frac{\cos\omega+j\sin\omega-\frac{1}{2}}{1-\frac{1}{2}\cos\omega-\frac{1}{2}j\sin\omega}$$

$$|H(e^{j\omega})| = \sqrt{\frac{\left(\cos\omega-\frac{1}{2}\right)^2+\sin^2\omega}{\left(1-\frac{1}{2}\cos\omega\right)^2+\frac{1}{4}\sin^2\omega}}$$

$$= \sqrt{\frac{\cos^2\omega+\frac{1}{4}-\cos\omega+\sin^2\omega}{1+\frac{1}{4}\cos^2\omega-\cos\omega+\frac{1}{4}\sin^2\omega}}$$

$$= 1$$

Problem 10 b

6.37. Consider a causal LTI system whose frequency response is given as:

$$H(e^{j\omega}) = e^{-j\omega}\frac{1-\frac{1}{2}e^{j\omega}}{1-\frac{1}{2}e^{-j\omega}}.$$

(a) Show that $|H(e^{j\omega})|$ is unity at all frequencies.

(b) Show that

$$\measuredangle H(e^{j\omega}) = -\omega - 2\tan^{-1}\left(\frac{\frac{1}{2}\sin\omega}{1-\frac{1}{2}\cos\omega}\right).$$

$$H(e^{j\omega}) = e^{-j\omega}\cdot\left(1-\frac{1}{2}e^{j\omega}\right)\cdot\frac{1}{1-\frac{1}{2}e^{-j\omega}}$$

1. $\angle e^{-j\omega} = -\omega$

2. $\angle\left(1-\frac{1}{2}e^{j\omega}\right) = \angle\left(1-\frac{1}{2}\cos\omega-\frac{1}{2}j\sin\omega\right) = -\tan^{-1}\left(\frac{\frac{1}{2}\sin\omega}{1-\frac{1}{2}\cos\omega}\right)$

3. $\angle\left(\frac{1}{1-\frac{1}{2}e^{-j\omega}}\right) = -\angle\left(1-\frac{1}{2}e^{-j\omega}\right) = -\angle\left(1-\frac{1}{2}\cos\omega+\frac{1}{2}j\sin\omega\right)$

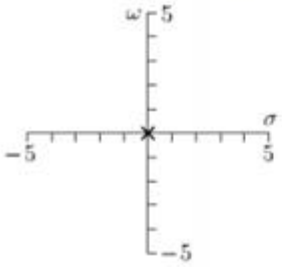
$$= -\tan^{-1}\left(\frac{\frac{1}{2}\sin\omega}{1-\frac{1}{2}\cos\omega}\right)$$

$$\angle H(e^{j\omega}) = -\omega - 2\tan^{-1}\left(\frac{\frac{1}{2}\sin\omega}{1-\frac{1}{2}\cos\omega}\right)$$

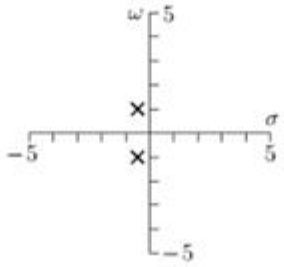
Problem 11

The following plots show pole-zero diagrams, impulse responses, Bode magnitude plots, and Bode angle plots for six causal CT LTI systems. Determine which corresponds to which.

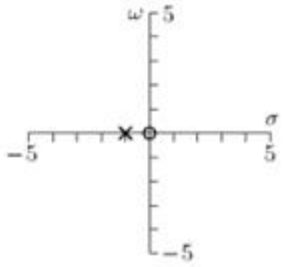
Pole-zero diagram 1



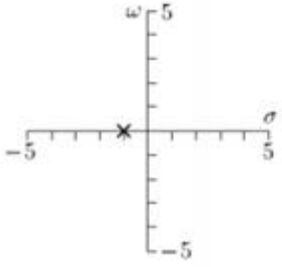
Pole-zero diagram 2



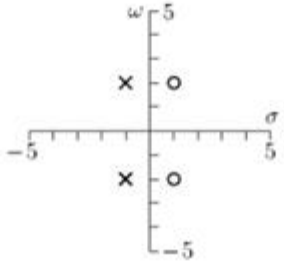
Pole-zero diagram 3



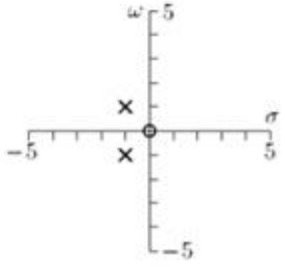
Pole-zero diagram 4



Pole-zero diagram 5

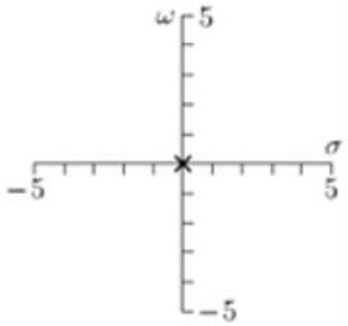


Pole-zero diagram 6

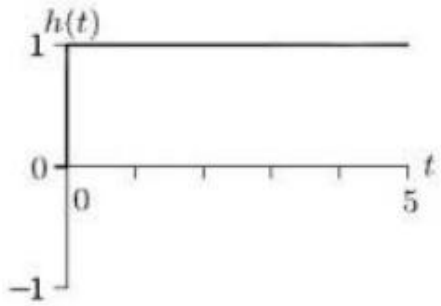


Problem 11-1

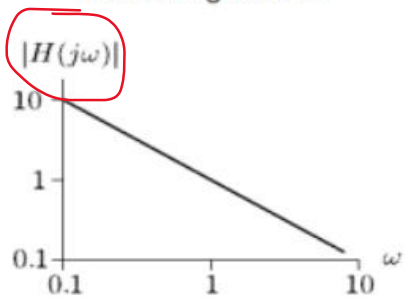
Pole-zero diagram 1



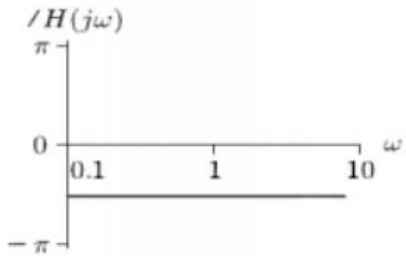
Impulse response 3



Bode Magnitude 5



Bode Angle 4



Bode Angle 2

$$p_0 = 0$$

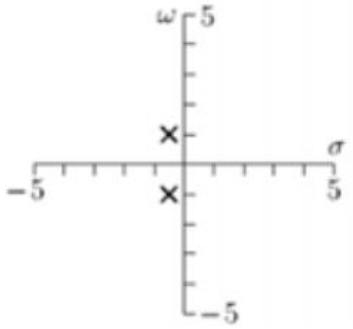
$$H(s) = \frac{1}{s}$$

$$h(t) = u(t)$$

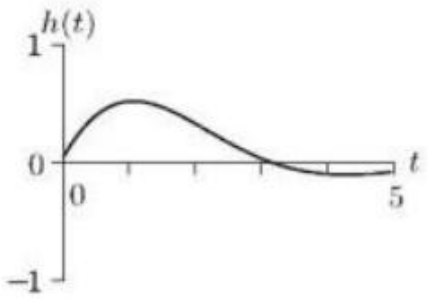
$$H(j\omega) = \frac{1}{j\omega}$$

Problem 11-2

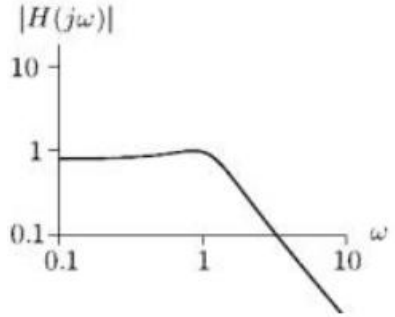
Pole-zero diagram 2



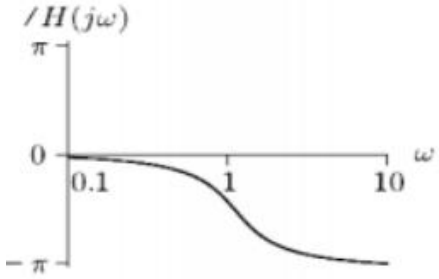
Impulse response 1



Bode Magnitude 2



Bode Angle 3



$p_0 = -m + j$

$p_1 = -m - j$

$0 < m < 1$

$[\sin \omega_0 t]u(t)$	$\frac{\omega_0}{s^2 + \omega_0^2}$	$\mathcal{R}e\{s\} > 0$	
Shifting in the *s*-Domain	$e^{s_0 t}x(t)$	$X(s - s_0)$	**Shifted version of *R* (i.e., *s* is in the ROC if $s - s_0$ is in R)**

$$H(s) = \frac{1}{(s+m)^2+1}$$

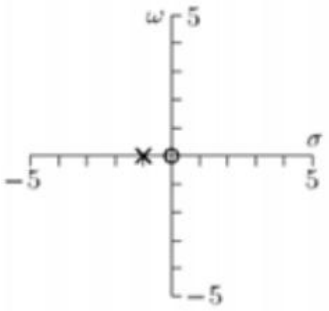
$$h(t) = e^{-mt}\sin t u(t)$$

$$H(j\omega) = \frac{1}{(j\omega+m)^2+1}$$

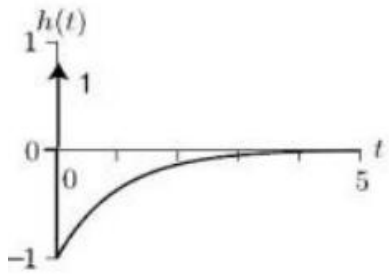
$$\omega_0 = 1$$

Problem 11-3

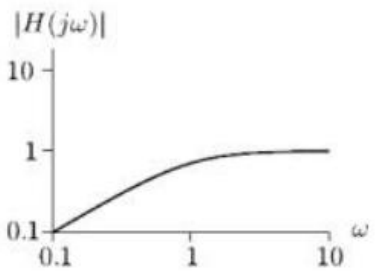
Pole-zero diagram 3



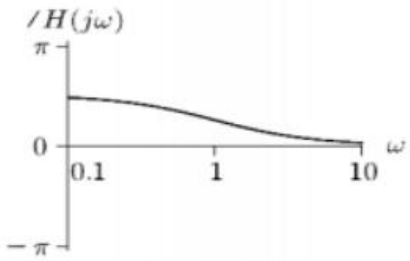
Impulse response 4



Bode Magnitude 3



Bode Angle 6



$z_0 = 0$

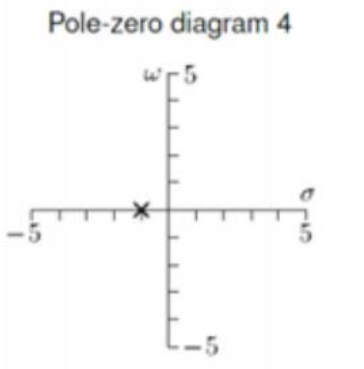
$p_0 = -1$

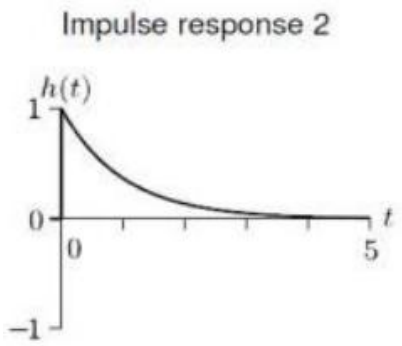
$$H(s) = \frac{s}{s+1} = 1 - \frac{1}{s+1}$$

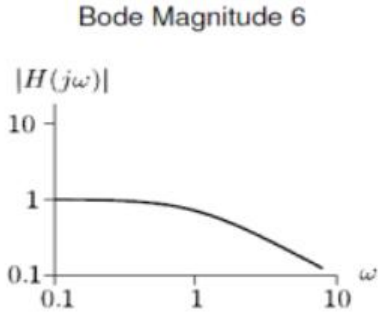
$$h(t) = \delta(t) - e^{-t}u(t)$$

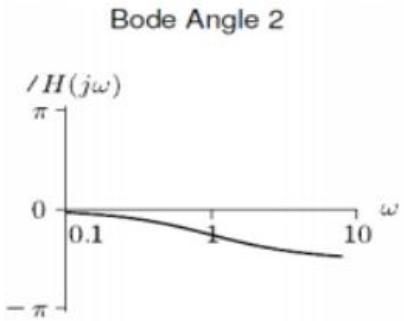
$$H(j\omega) = \frac{j\omega}{j\omega + 1}$$

Problem 11-4









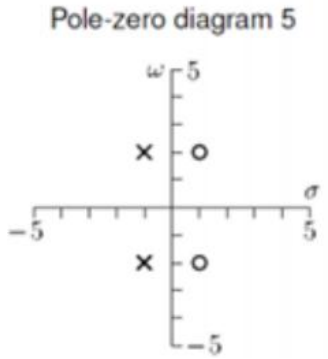
$$p_0 = -1$$

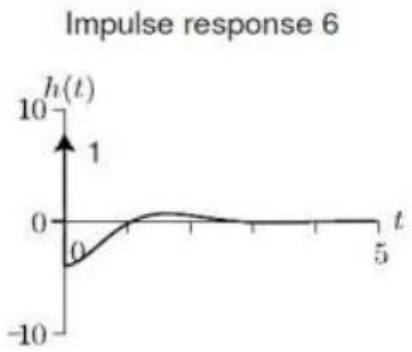
$$H(s) = \frac{1}{s+1}$$

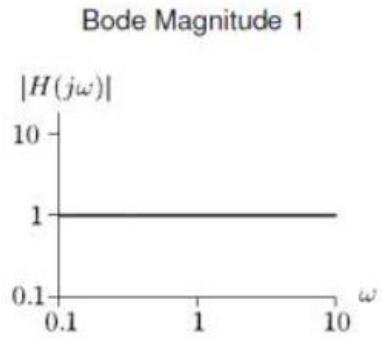
$$h(t) = e^{-t}u(t)$$

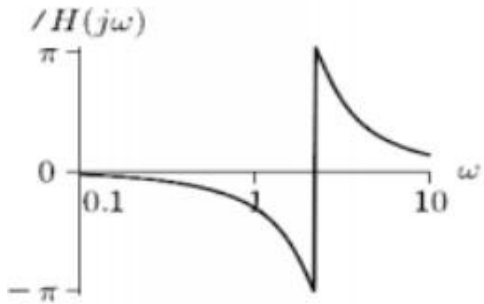
$$H(j\omega) = \frac{1}{j\omega + 1}$$

Problem 11-5









$z_0 = 1 + j2, \quad z_1 = 1 - j2$

$p_0 = -1 + j2, \quad p_1 = -1 - j2$

$$H(s) = \frac{(s-1)^2 + 4}{(s+1)^2 + 4}$$

$$= 1 - \frac{4s}{(s+1)^2 + 4}$$

$$= 1 - \frac{4(s+1)}{(s+1)^2 + 4} + \frac{4}{(s+1)^2 + 4}$$

$[\cos \omega_0 t]u(t)$	$\frac{s}{s^2 + \omega_0^2}$	$\mathcal{R}e\{s\} > 0$
$[\sin \omega_0 t]u(t)$	$\frac{\omega_0}{s^2 + \omega_0^2}$	$\mathcal{R}e\{s\} > 0$

$$h(t) = \delta(t) - 4e^{-t}\cos(2t)u(t) + 2e^{-t}\sin(2t)u(t)$$

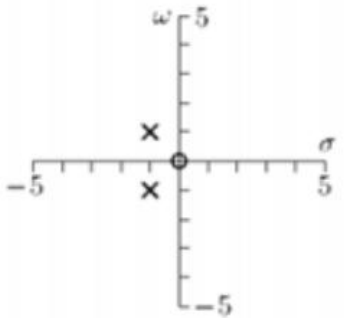
$$= \delta(t) + Ae^{-t}\cos(2t + \phi)u(t)$$

$$H(j\omega) = \frac{(j\omega - 1)^2 + 4}{(j\omega + 1)^2 + 4}$$

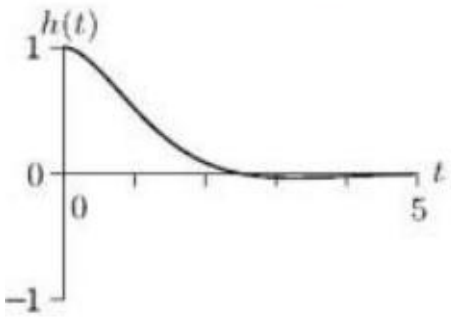
$$\omega_0 = \sqrt{5}$$

Problem 11-6

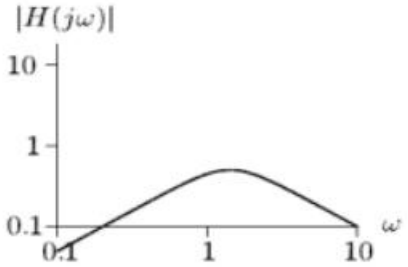
Pole-zero diagram 6



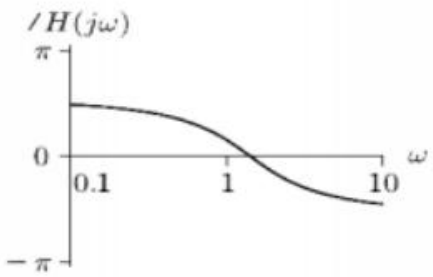
Impulse response 5



Bode Magnitude 4



Bode Angle 5



$$z_0 = 0$$

$$p_0 = -1 + j,\ \ p_1 = -1 - j$$

$$H(s) = \frac{s}{(s+1)^2 + 1}$$

$$= \frac{s+1}{(s+1)^2 + 1} - \frac{1}{(s+1)^2 + 1}$$

$$h(t) = e^{-t}\cos t u(t) - e^{-t}\sin t u(t)$$

$$= \sqrt{2}e^{-t}\cos\left(t - \frac{\pi}{4}\right)u(t)$$

$$H(j\omega) = \frac{j\omega}{(j\omega+1)^2+1}$$

$$\omega_0 = \sqrt{2}$$